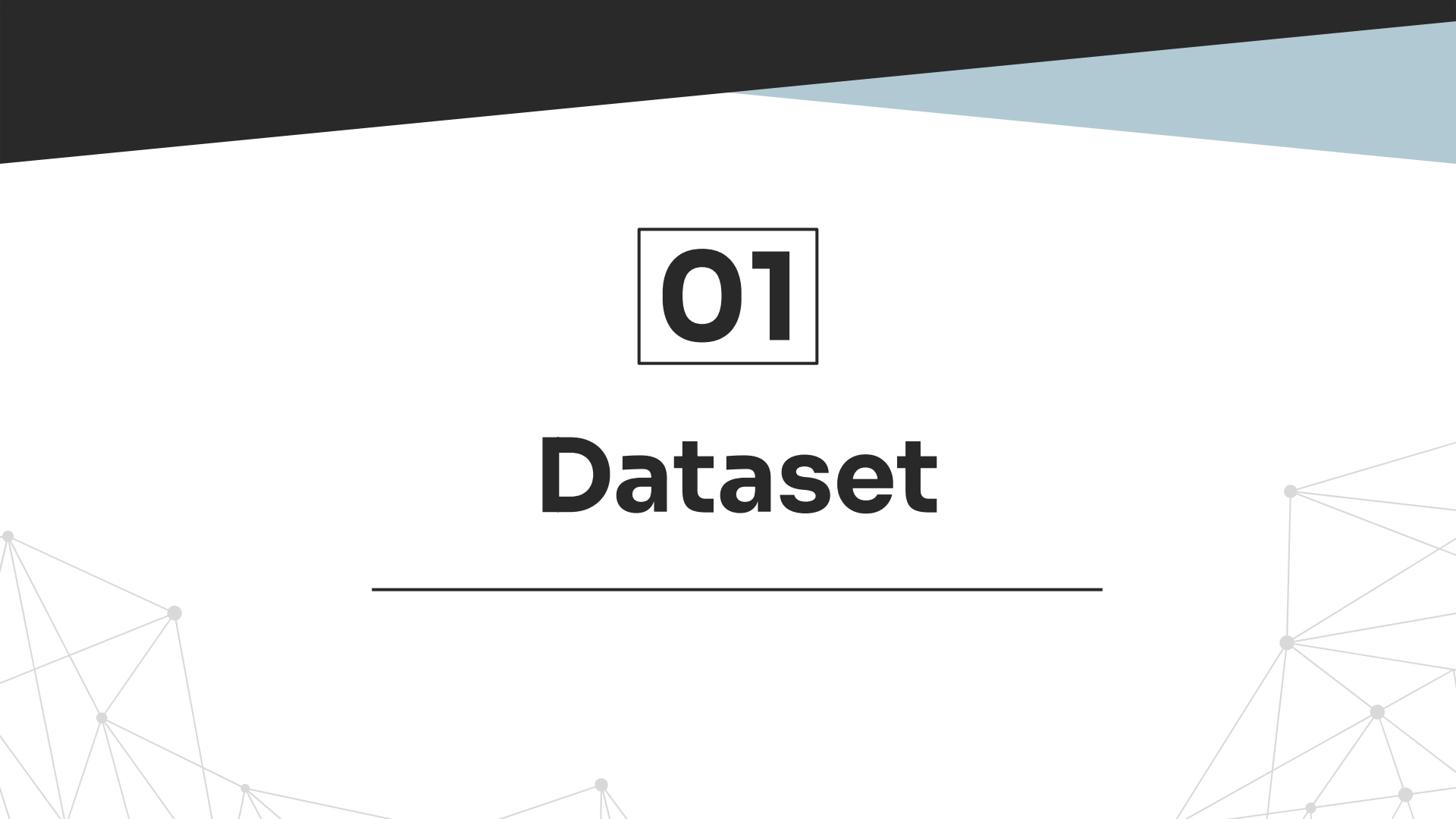




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Noise – Robust Face Representations : Comparing Human **ECoG** to **CNN** Representations of Noisy Faces

NoiSix: Jana, Mariana Nicoli, Rodrigo Castillo,
Alexandra Sheridan, Elide Portocarrero

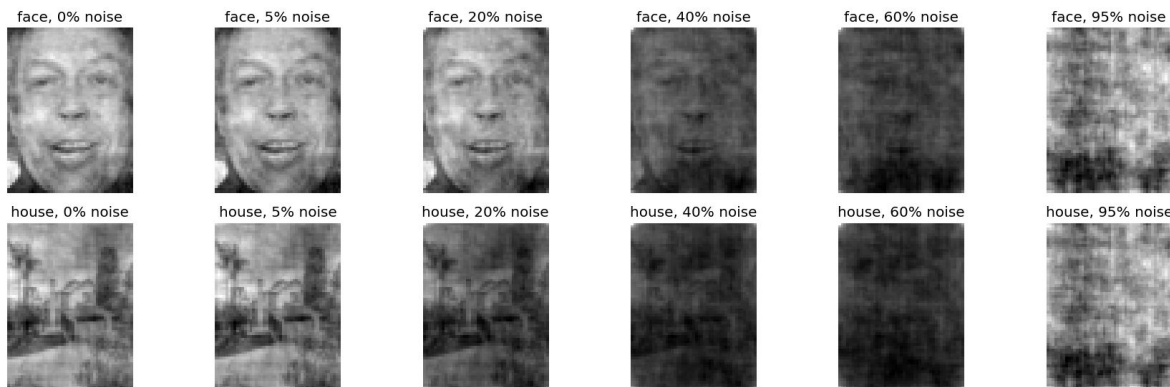


01

Dataset

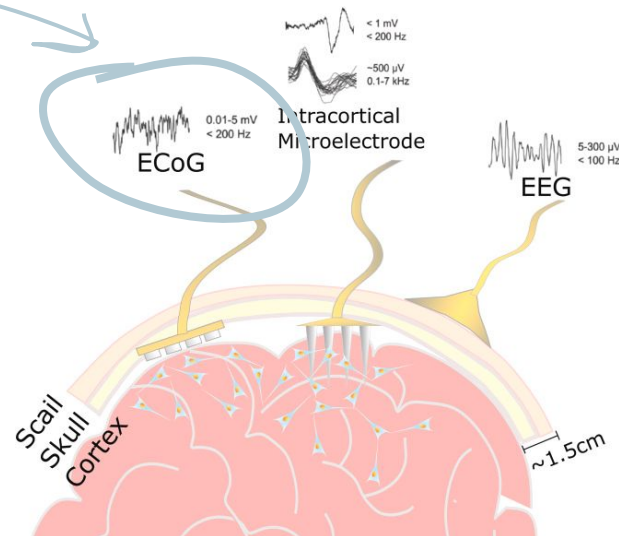
Dataset & Task: Miller 2017

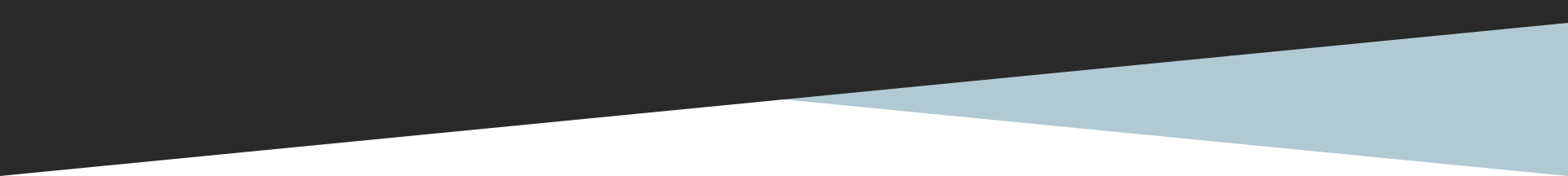
- Noise via phase-scrambling: FFT
- 0–100% noise, in 5% steps
- Recombined with average magnitude → all stimuli share the same power spectrum (only noise varies)



ted8080. (2019). House Prices and Images - SoCal [Data set]. Kaggle.
www.kaggle.com/datasets/ted8080/house-prices-and-images-socal

- ECoG data
- 5 epileptic patients
- Electrodes in face-selective regions
- 38 base images (faces + houses)





02

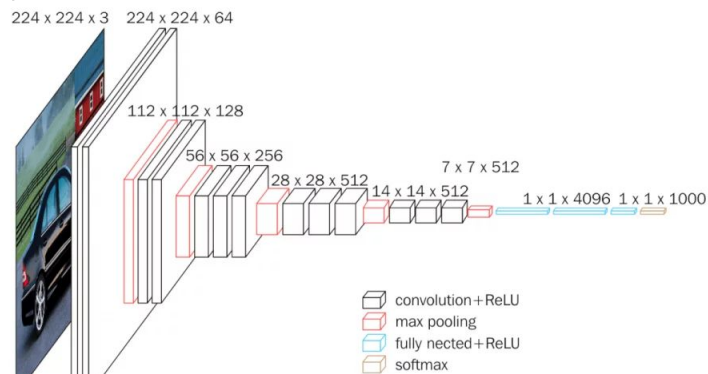
Methods & Results



1- Group Electrodes & CNN Layers

VGG-16 trained on ImageNet

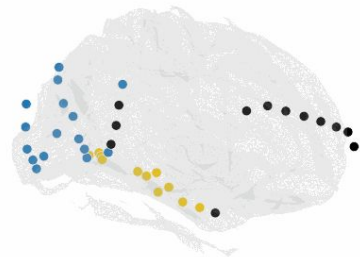
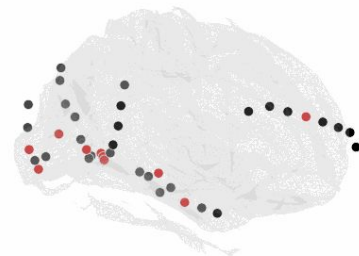
Pooling Layers, Fully Connected Layers



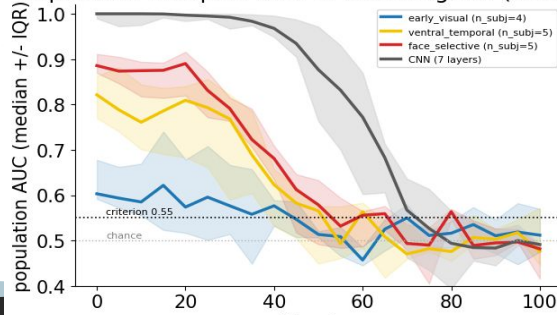
Added Noise to the CNN Units
Fine-tuned CNN for Face Detection

ECoG Electrodes: 3 groups

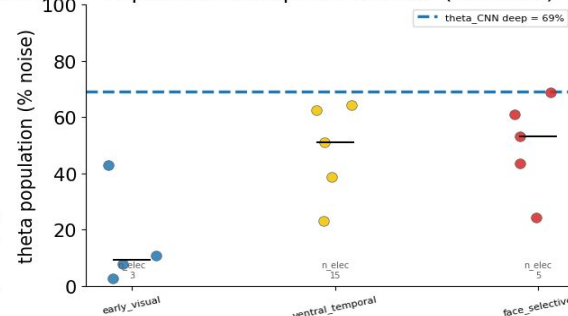
Early Visual, Ventral Temporal, Face Selective



Population collapse: CNN vs ECoG regions (crit 0.55)

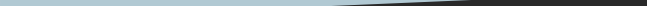


Population collapse threshold (crit 0.55)

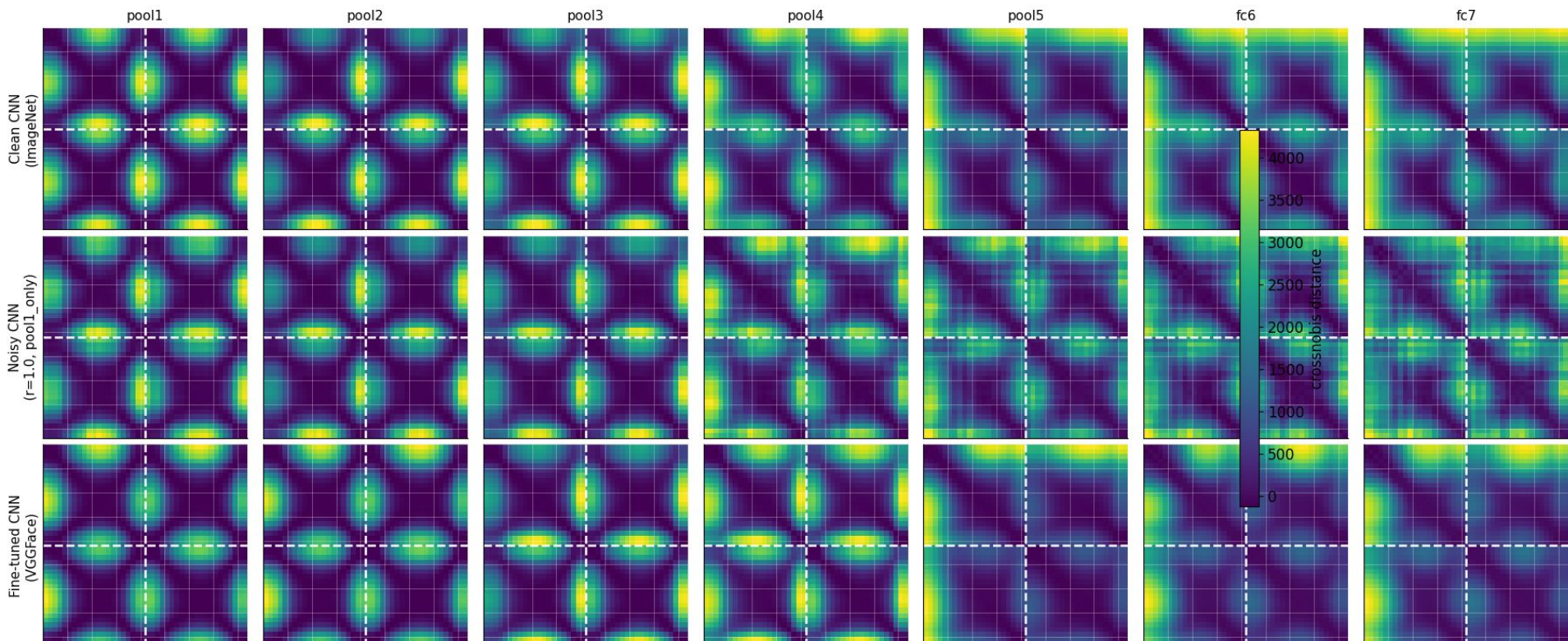




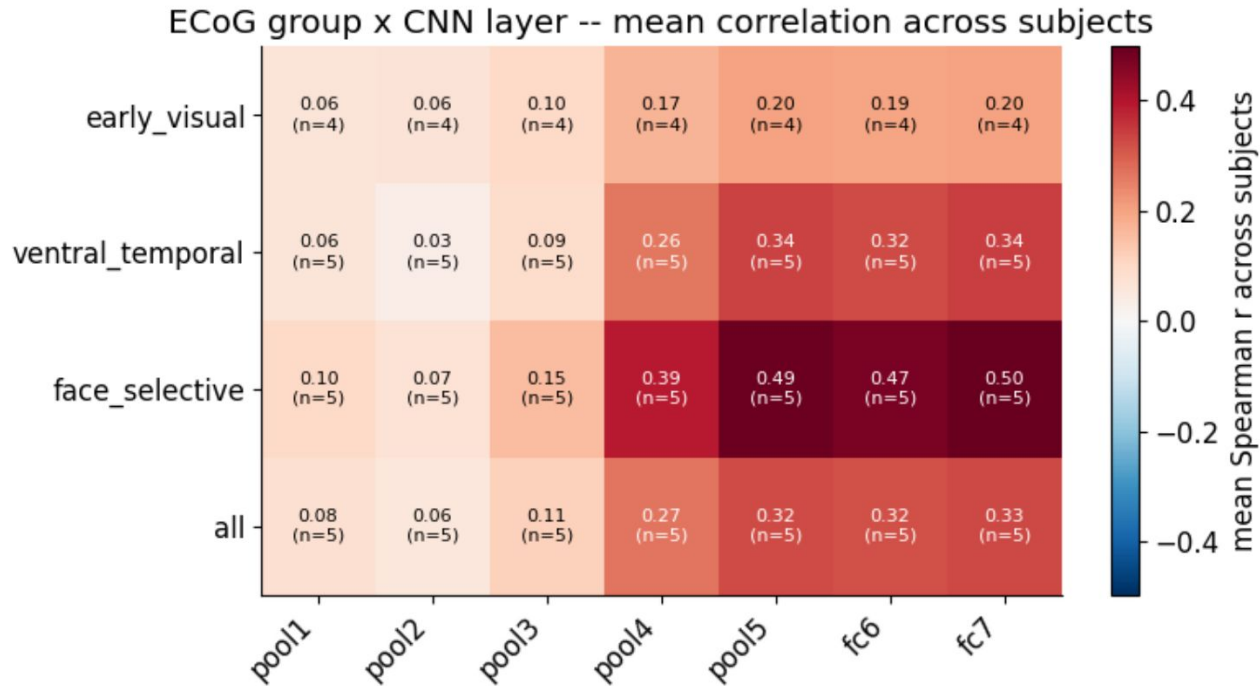
2- Build ECoG RDMs



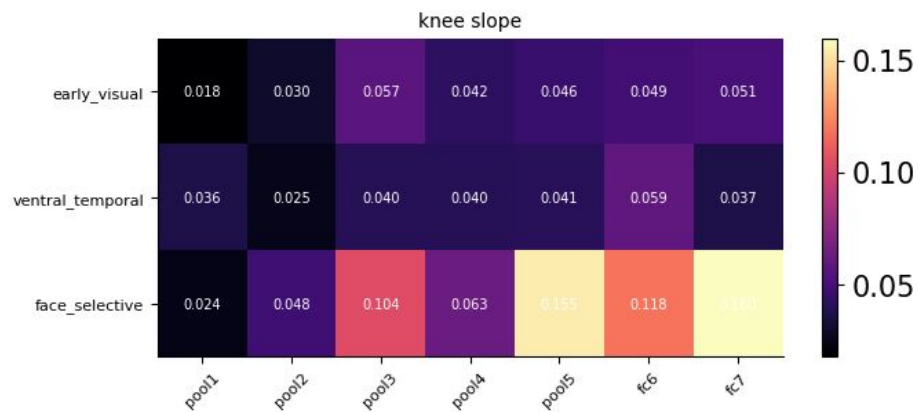
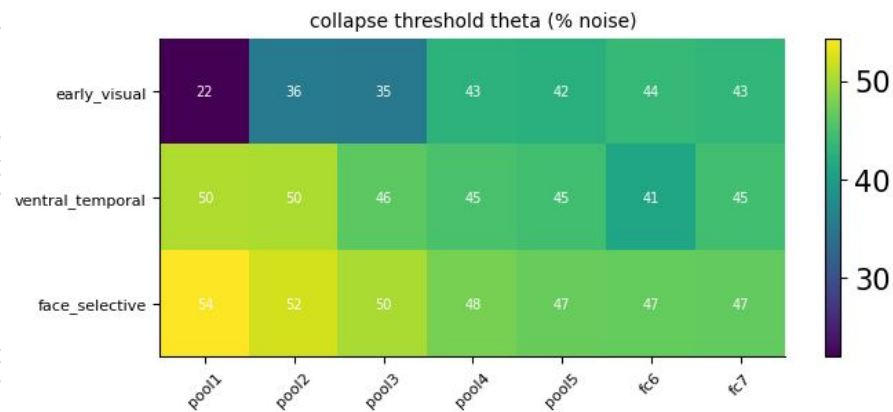
RDMs by CNN type x layer -- category-major ordering (face block | house block)



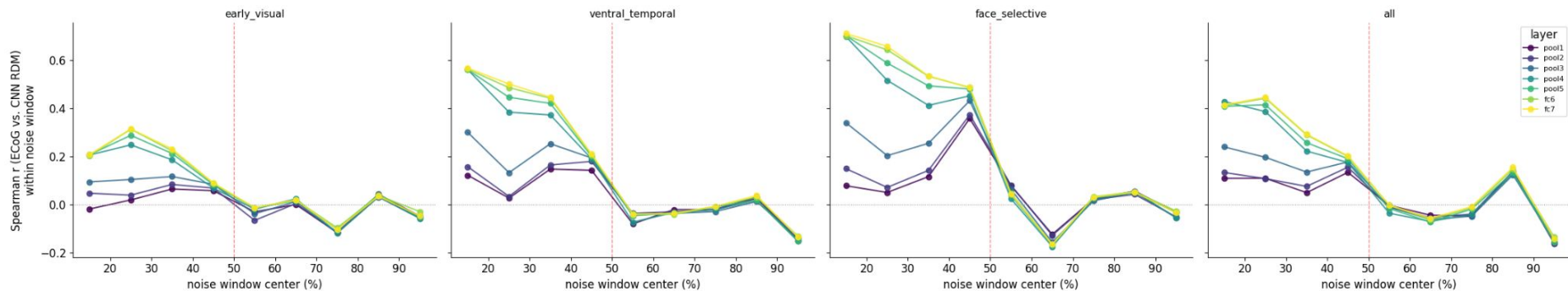
CNN x ECoG Correlation



Correspondence collapse: ECoG group x CNN layer

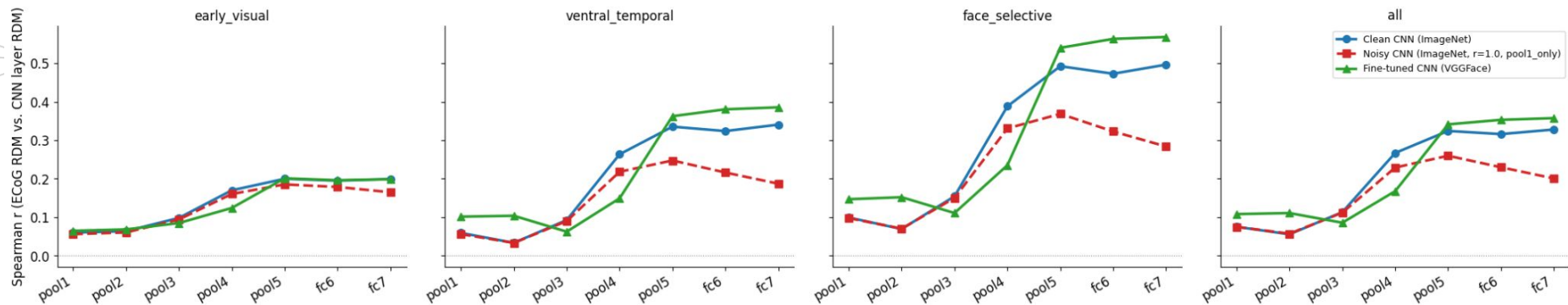


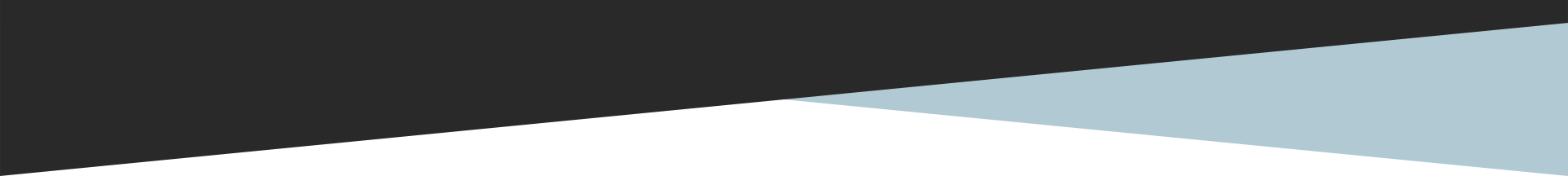
RDM correspondence vs. noise level (mean across subjects; red = 50% reference)



Adding Noise and Fine Tuning

ECoG x CNN correspondence: clean vs. noisy vs. fine-tuned CNN



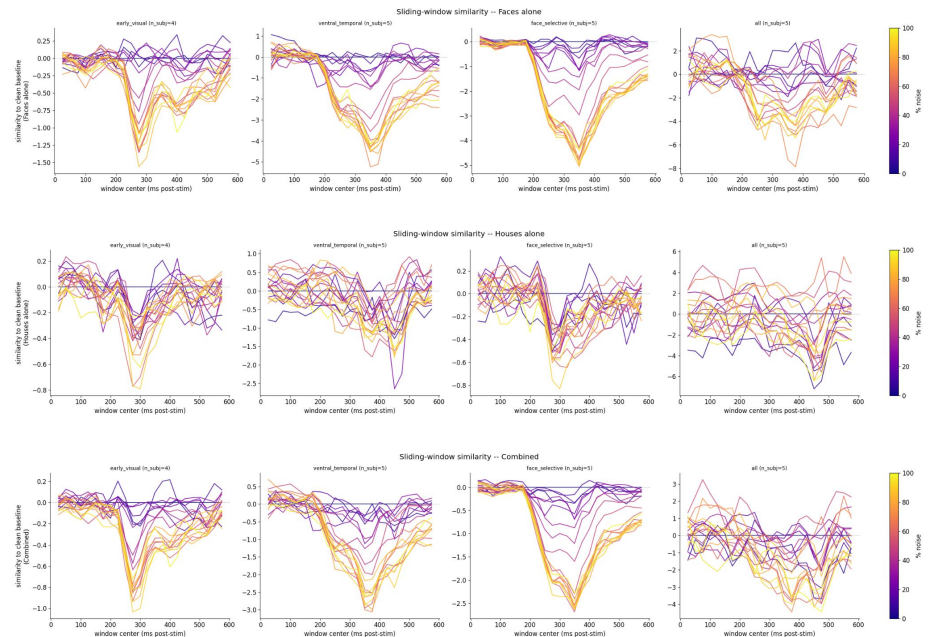
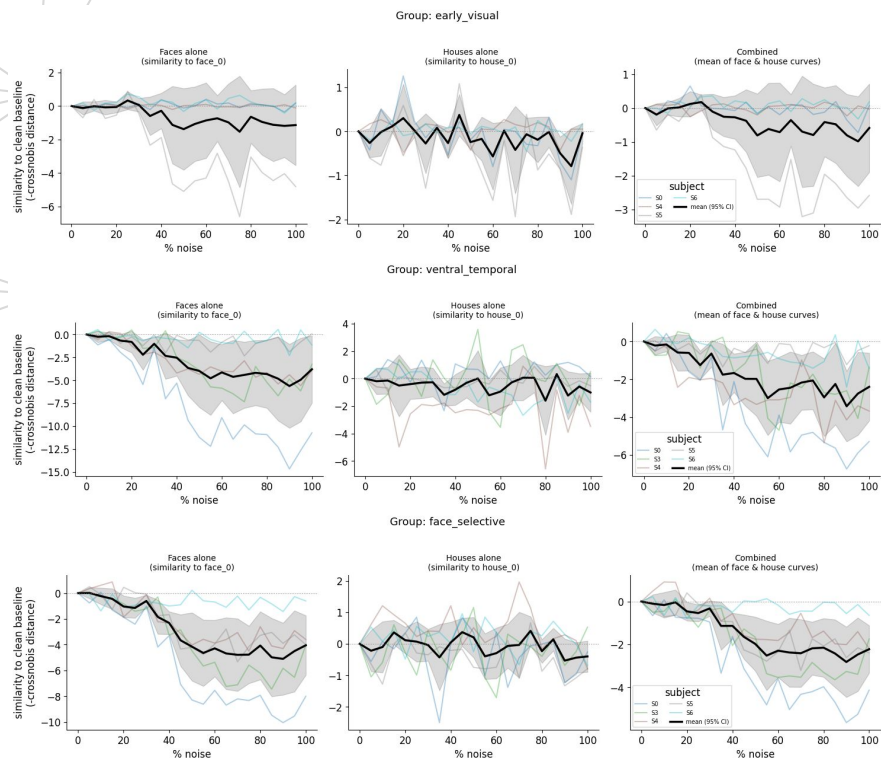


03

Supplementary Figures

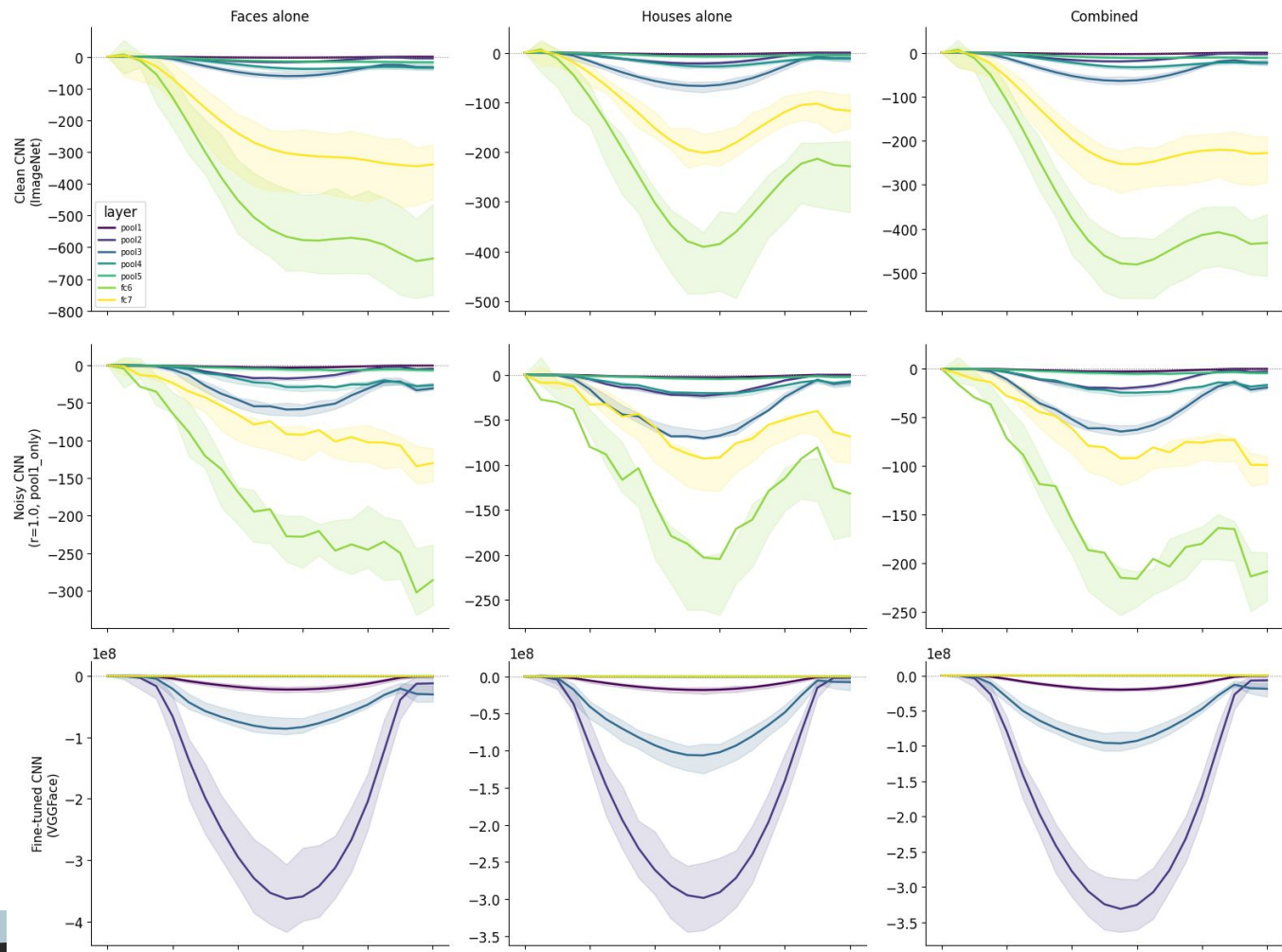


ECoG Similarity to Clean baseline vs. Noise



Similarity to clean baseline vs. noise -- clean / noisy / fine-tuned CNN, all layers
line=primary, band=bootstrap 95% CI

CNN Similarity to Clean Baseline vs. Noise





Thank You! Questions?

In Conclusion,

- CNN representations increasingly resemble face-selective and ventral-temporal cortex with depth, collapsing near the human perceptual threshold (~45–50% noise).
- Early visual cortex shows weak, shallow correspondence throughout — it simply isn't doing face-specific work.
- Fine-tuning for faces strengthens this brain–CNN correspondence; injecting noise weakens it — the CNN's *training objective*, not just its architecture, shapes how brain-like its noisy-face representations are

Full Project and Code Available on [GitHub](#)

